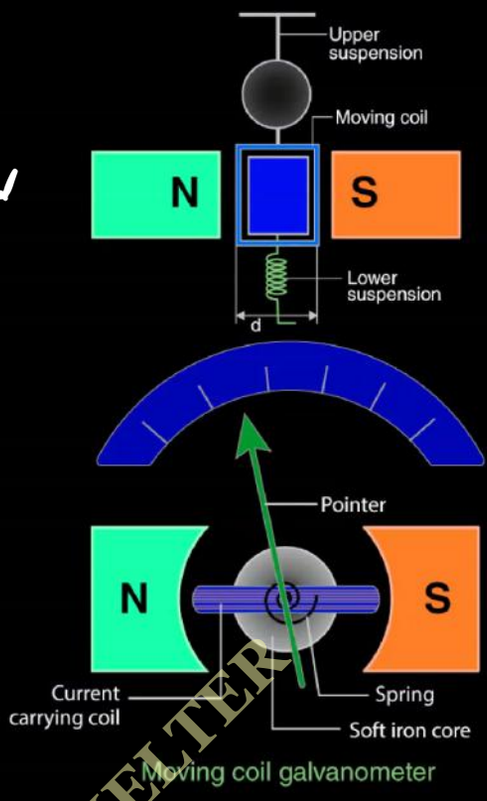
MOVING COIL GALVANOMETER: It is a device for measuring small currents. It comprises of a coil having some resistance and a connected pointer which can move on a etched scale. It is based on principle of magnetic torque whose mechanism we will learn in the chapter on magnetics. A Galvanometer is characterized by following two quantities:

1. Coil Resistance (G)
2. Full Scale deflection Current i.e the current at which maximum possible deflection of the Galvanometer needle occurs.

→ When no current flows through a Galvanometer, the needle remains at the centre and deflects to right or left depending on the direction of current flow through its coil.

→ Typical range of a Galvanometer is in mA.

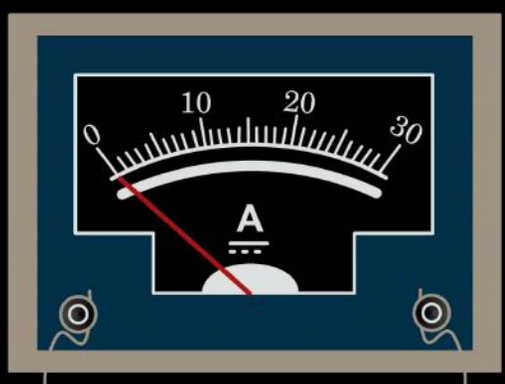


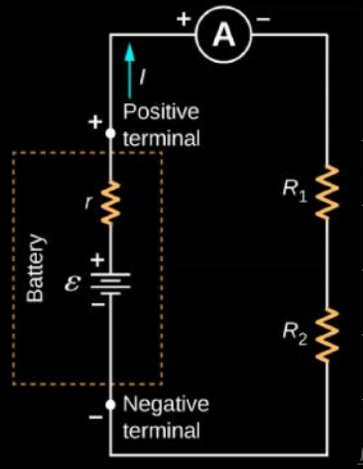


AMMETER: It is a device used for measuring relatively larger currents (Typically between 1 to 50 Amperes however could be larger also). An ammeter can be obtained by suitably modifying a Galvanometer (Will be explained in next section).

→ For measuring current, an Ammeter is connected in series with branch whose current is to be measured.

→ An ammeter should have very low resistance (typically fraction of a milli ohm, otherwise it will significantly change the effective resistance of





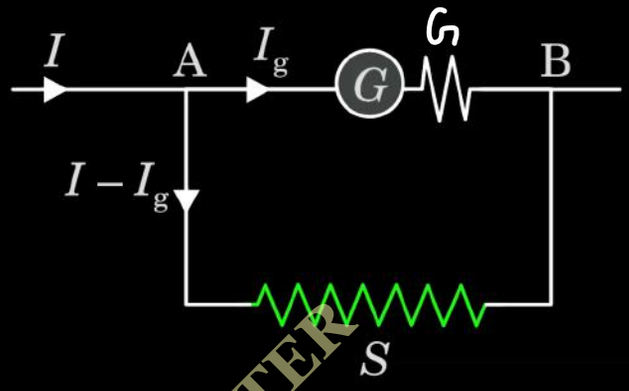
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circuit considerably and thus change the current being measured.

→ Ideal ammeter has a zero resistance.

Converting A Galvanometer To Ammeter:

A Galvanometer can measure only small currents however it can be converted to measure large currents by attaching a "shunt" or a parallel resistor (parallel to its own coil). Suppose the full scale deflection current of Galvanometer is I_g and we wish to convert it to a range I (i.e when a current I flows in the main branch, we wish Galvanometer to show full scale deflection) then we can find the required value of shunt using following equation (Refer to figure):



$(I - I_g)S = I_g G$, Here G is coil resistance of Galvanometer.

Thus, $\boxed{S = \left(\frac{I_g}{I - I_g}\right) G}$, Now whatever current is flowing through Galvanometer, we can scale it by a factor (I/I_g) to get actual current in the main branch.

→ Effective Resistance of Ammeter thus formed is $\boxed{R_A = \frac{GS}{G+S}}$

→ Calibration factor: We must multiply the Galvanometer reading by I/I_g, this is known as calibration factor. Rearranging the equation for finding shunt, we get $I = \left(\frac{G+S}{S}\right) I_g$ thus our calibration factor is $(G+S)/(S)$. i.e $\boxed{k = I/I_g = (G+S)/S}$

→ Most of the times $I_g << I$ and we might approximate $I - I_g \approx I$.

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EXAMPLE: A moving coil Galvanometer of resistance of 100Ω is used as an ammeter using a resistance of 0.1Ω. Maximum deflection current of Galvanometer is 100μA. Find the range of the ammeter thus formed.

Solution: $I = \left(\frac{G+S}{S}\right) I_g = \left(\frac{100+0.1}{0.1}\right) \times 100\,\mu A = 100.1\,mA$ Ans.

EXAMPLE: A Galvanometer of resistance 20Ω gives a full scale deflection at 0.04 A. It is desired to convert it to ammeter of range 20 A. The only available shunt is 0.05Ω. What resistance should be added to Galvanometer coil in series.

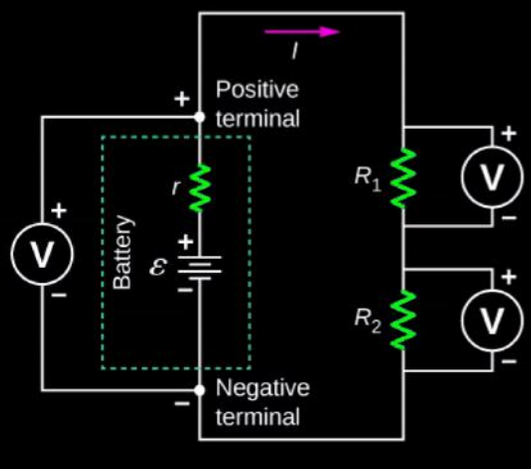
Solution: $I = I_g \frac{(G_{eff}+S)}{S}$ $(G_{eff} = G + R_{series})$

$\Rightarrow G_{eff} = \left(\frac{I}{I_g} - 1\right) S = \left(\frac{20}{0.04} - 1\right)(0.05) = 24.95\Omega$

$\Rightarrow$ Required series resistance $R = G_{eff} - G = 4.95\Omega$ Ans.

VOLTMETER: A voltmeter is an instrument to measure potential difference between any two points in a circuit. A voltmeter is always connected in parallel to the branch across which we wish to find the potential difference (see the figure below). A Voltmeter should have a very large resistance (Typical Voltmeters have resistance of order of MΩ while high precision Voltmeters may even have a resistance of order of 10 GΩ).





→ An ideal voltmeter has infinite resistance.

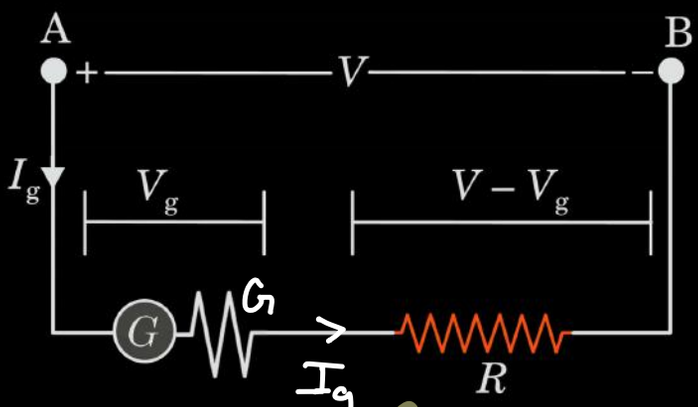
→ A voltmeter can be made by converting a Galvanometer using a high series resistance

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this will be explained in next section.

Converting A Galvanometer To Voltmeter:

Suppose we have a Galvanometer of full scale deflection current I_g and coil resistance G. Then for full scale deflection, potential drop across its coil is given as $V_g = I_g G$. Now if we wish to increase it range to V, we need to add a series resistance R (Refer to adjacent figure). $I_g (G + R) = V \Rightarrow R = \frac{V}{I_G} - G = \frac{V}{V_g} G - G$



$\Rightarrow$ $\boxed{R = G\left(\frac{V}{V_g} - 1\right)}$, we can also rearrange this equation as: $\boxed{V = \left(\frac{G+R}{G}\right) V_g}$

→ Sensitivity of any instrument is defined as change in output to the change in input signal (for instance for voltmeter it will be change in angle of deflection per unit change in voltage i.e sensitivity will be measured in divisions per volt). Sensitivity and range are inversely related. As the range is increased, sensitivity will reduce.

Example: A Galvanometer has coil resistance 20 Ω and full scale deflection current 1.0 mA. How will you convert it voltmeter of range 10V?

Solution: $R = G\left(\frac{V}{V_g} - 1\right) = 20\left[\frac{10}{20 \times 1 \times 10^{-3}} - 1\right] = 9980\ \Omega$

Thus we will need to connect a series resistance of 9980 Ω. Ans.

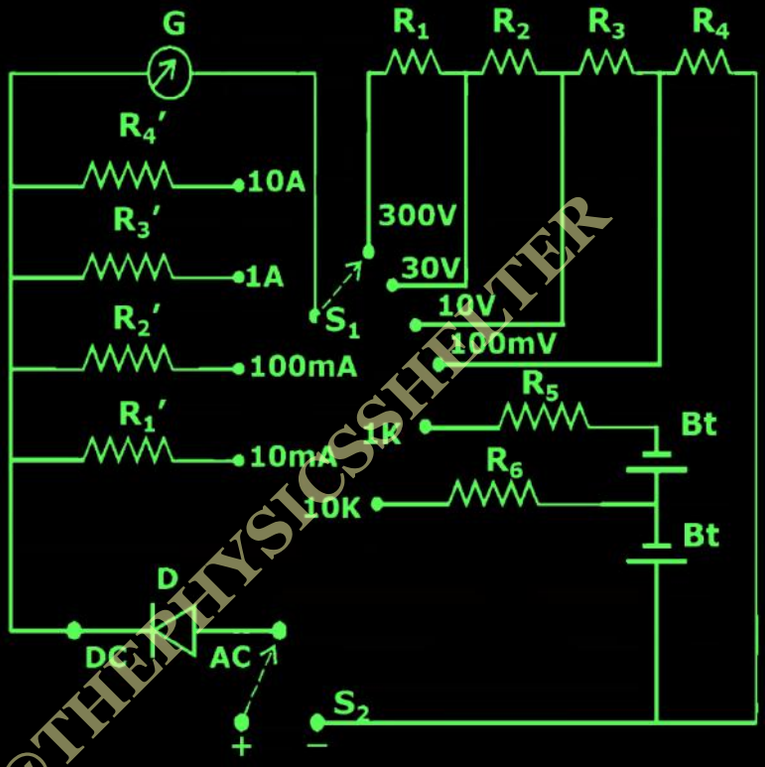
Ohm Metre: An ohm-metre is used to find the resistance of any circuit component or device. Traditional ohm-metres contained an internal voltage source (battery) that would be connected across the component being tested. A current is generated in

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the component which is measured using a Galvanometer and then ohm's law is used to deduce the resistance. Nowadays digital meters are used which pass a constant current through the component and the voltage is measured and calibrated against the resistance.

MULTIMETER:





A multimeter is a compact device which can easily be converted to a Voltmeter, Ammeter or an ohmmeter by suitably turning its knob to a desired location. By turning the knob, various shunts, Series resistances or batteries automatically get connected to its Galvanometer which is read out for finding the appropriate physical quantity. A sample internal circuitry of the multimeter is shown above and one can get an idea about how switching process modifies the effective circuit.

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METRE BRIDGE:

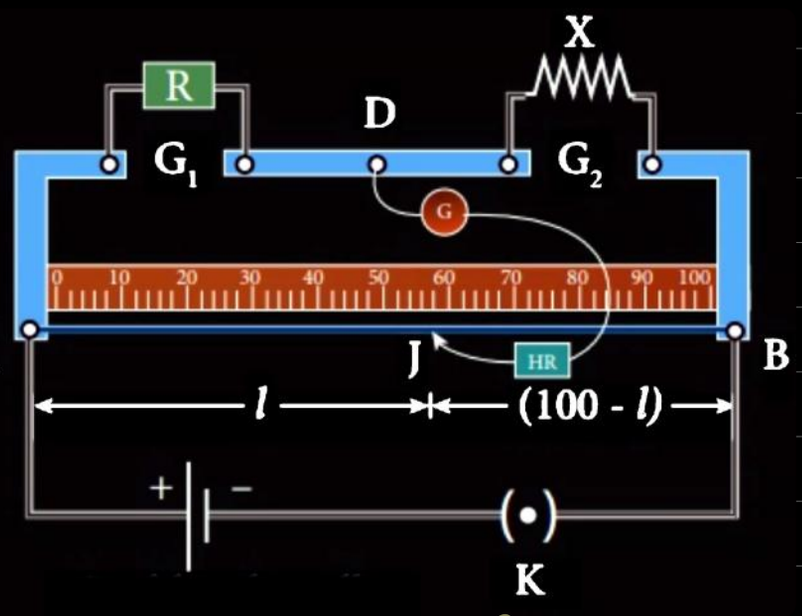
It is a device based on principle of balanced wheatstone bridge which can be used for estimating unknown resistances. It consists of a 1metre long wire and two gaps between highly conducting metallic strips between which various known or unknown resistors can be connected. The schematic equivalent circuit of a metre bridge is shown in the adjacent figure. For Galvanometer to show zero reading,

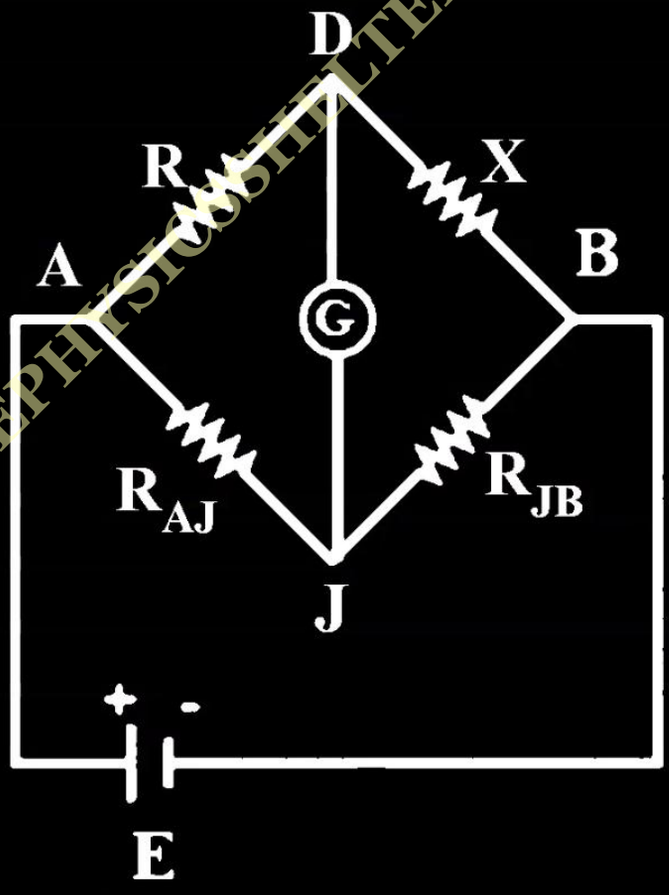
$$\frac{R}{X} = \frac{R_{AJ}}{R_{JB}}$$, however since

the wire on which Jockey slides is a uniform wire thus R_{AJ} and R_{JB} are simply proportional to lengths of corresponding sections thus, $\frac{R}{X} = \frac{l}{100-l}$ (l in cm)

Thus unknown resistance X is estimated as :

$$X = \left(\frac{100-l}{l}\right) R$$





→ In this analysis we assumed all other resistances except R, X and R_{AJ} and R_{JB} to be zero however the conducting metallic strips and the points of wire connections will also have some effective resistances. For an accurate analysis, these factors are accounted by considering end corrections i.e substituting $l+\alpha$ for l and $100-l+\beta$ for $100-l$ where α and β are considered end corrections for left end and right ends in cm.

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→ To minimize error in calculation of X, R should be chosen so as to get the null point around middle of the Galvanometer wire, this can be seen from error theory:

$$X = \frac{R\,(100-\ell)}{\ell} \Rightarrow \ln X = \ln(100-\ell) - \ln \ell + \ln R$$

$$\Rightarrow \frac{dX}{X} = \left(\frac{1}{100-\ell} - \frac{1}{\ell}\right) d\ell \neq 0 \quad \text{(If } R \text{ is known exactly)}$$

Thus for $dX = 0$, $\frac{1}{\ell} = \frac{1}{100-\ell} \Rightarrow \ell = 50$ cm. (Here $d\ell$ represents least count of the the metre scale).

→ Sometimes to get a better estimate of X, the experiment is repeated by exchanging positions of X and R.

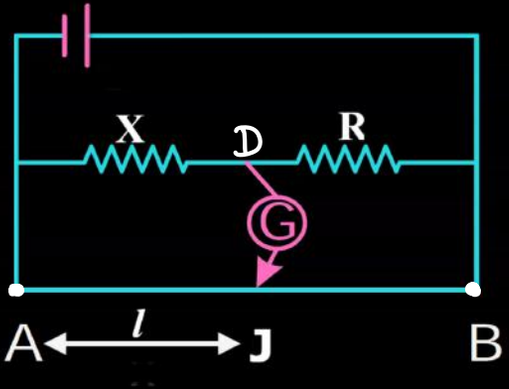
EXAMPLE:

A resistance of 2Ω is connected across one gap of a metre-bridge (the length of the wire is 100 cm) and an unknown resistance, greater than 2Ω, is connected across the other gap. When these resistances are interchanged, the balance point shifts by 20 cm. Neglecting any corrections, what is the unknown resistance.

Solution: $R = 2\Omega$ let initial balancing length be ℓ then upon exchanging the positions, balancing length must be $100-\ell$ and $\ell > 50$ cm ($\because X > R$). Thus according to question $\ell - (100-\ell) = 20$ cm

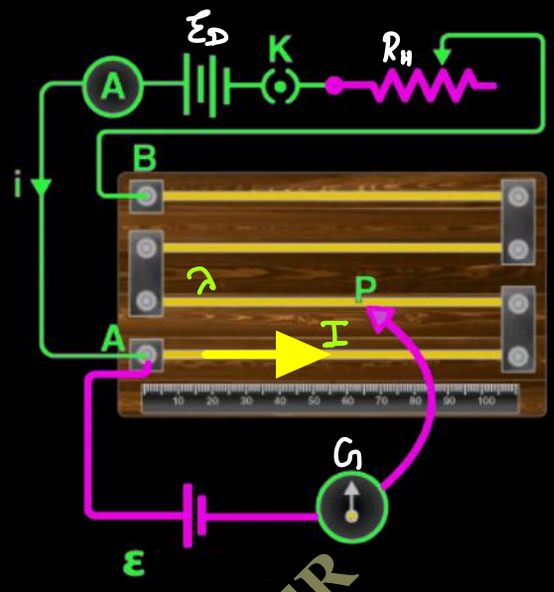
$\Rightarrow \ell = 60$ cm. Thus $\frac{X}{R} = \frac{60}{40} \Rightarrow X = 1.5R = 3\Omega$ Ans.

(See adjacent figure for more clarity, Here distances are measured from left end, thus $\because X > R$, $\ell > 50$ cm).



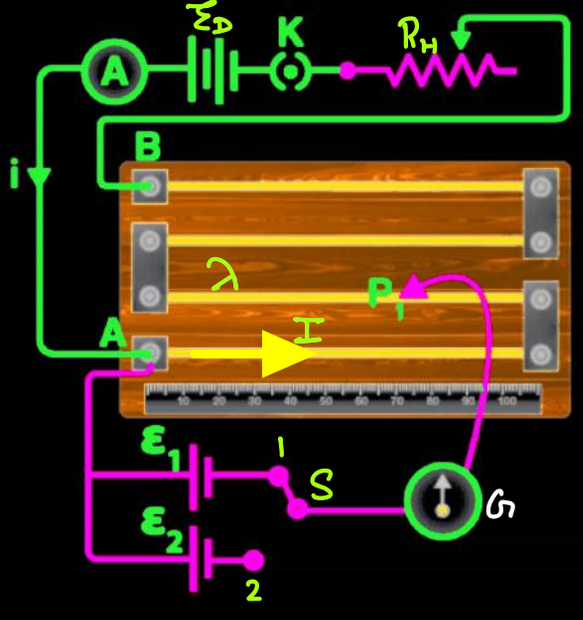
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POTENTIOMETER: It is essentially a long wire (usually made of nichrome). It can be used for comparing EMFs of two cells or finding internal resistance of a battery. A schematic of potentiometer set up is shown in adjacent figure. Here, $\mathcal{E}_D$ is a driver cell of sufficiently high voltage which is connected to a potentiometer wire through a Rheostat, key and an ammeter. The lower circuit (shown in magenta) has a Galvanometer which has a Jockey at the other end. The driver drives a current through the potentiometer wire (Yellow), and Jockey is placed such that there is zero current through Galvanometer. At this configuration, potential drop from A to P along Yellow wire must be equal to potential drop along path AGP (i.e EMF of the battery). Let I be the current in potentiometer wire at this configuration and let λ be the resistance per unit length of the wire and l the length of potentiometer wire from A to P then, $\mathcal{E} = I\lambda l$. The length of the potentiometer wire till the null point can be read using a scale placed on a wooden box on which the potentiometer wire is fixed.



USING POTENTIOMETER FOR COMPARING EMFS

For comparing EMFs of two cells, the circuit is configured as shown in the figure. First switch is thrown to position 1 and length for null point (say l_1) is noted. Thus as discussed in previous section, $\mathcal{E}_1 = I\lambda l_1$, similarly, when switch is thrown to position 2, let l_2 be the length for null point then $\mathcal{E}_2 = I\lambda l_2$.

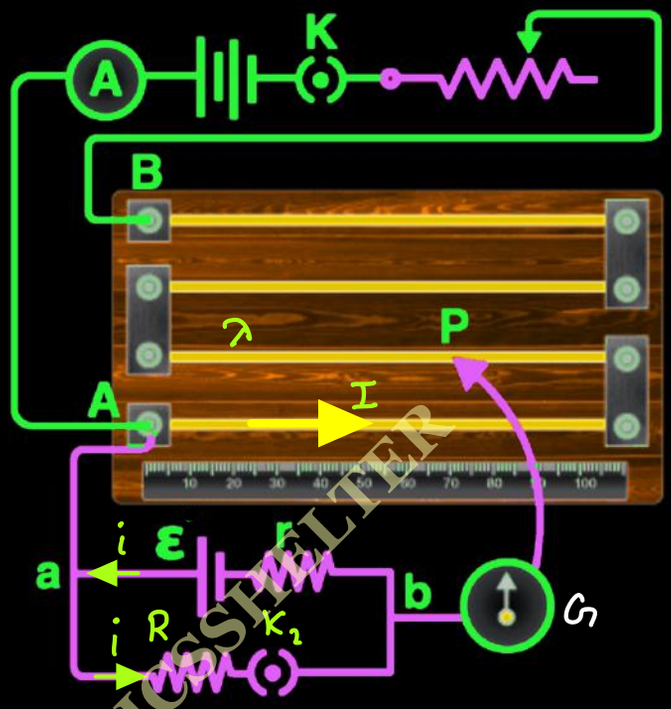


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Thus EMFs of two cells is simply compared as $\boxed{\frac{\mathcal{E}_1}{\mathcal{E}_2} = \frac{l_1}{l_2}}$

Using Potentiometer For Finding Internal Resistance:

For finding internal resistance of a cell, the potentiometer set up is configured as shown. First balance length (for null point) is found with K_2 pulled out (Thus R is in open circuit) let balance length obtained in this case be l_1 then, $\mathcal{E} = I\lambda l_1$ ①



Now the key k_2 is inserted and balance length is again found. In this case let i be the current in lower circuit (magenta) then at null point, $i = \frac{\mathcal{E}}{r+R}$ and potential drop from a to b is

$$V = \mathcal{E} - ir = \mathcal{E} - \frac{\mathcal{E}}{R+r}r = \mathcal{E}\left[\frac{R}{R+r}\right]$$, now let balancing length be l_2 then, $\mathcal{E}\left[\frac{R}{R+r}\right] = I\lambda l_2$ ②

Now from equations ①/② gives : $\frac{R+r}{R} = \frac{l_1}{l_2}$

$$\Rightarrow \boxed{r = R\left(\frac{l_1 - l_2}{l_2}\right)}$$

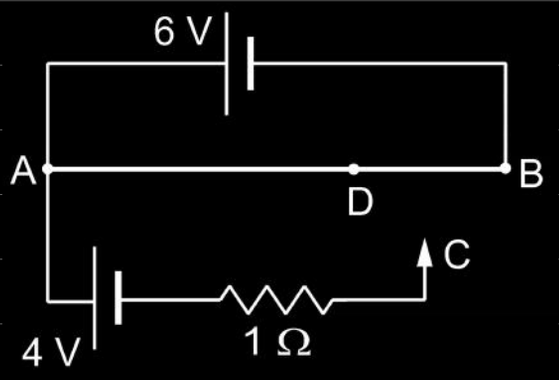
→ Total Potential drop across potentiometer wire must be greater than $\mathcal{E}$ otherwise null point will not exist.

→ Thus $\mathcal{E}_D$ must be greater than $\mathcal{E}$, though it is not a sufficient condition because there will also some potential drop across rheostat proportional to its resistance thus actually, $\mathcal{E}_D \times \frac{R_P}{R_H + R_P} > \mathcal{E}$ where R_P is resistance of potentiometer wire and R_H is resistance of the rheostat.

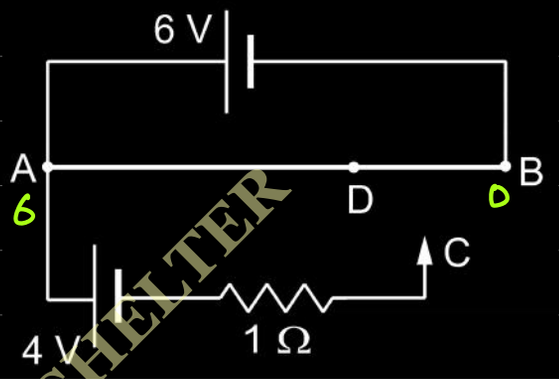
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Example: (H.C. Verma 55)

A 6-volt battery of negligible internal resistance is connected across a uniform wire AB of length 100 cm. The positive terminal of another battery of emf 4 V and internal resistance 1 Ω is joined to the point A as shown in figure. Take the potential at B to be zero. (a) What are the potentials at the points A and C? (b) At which point D of the wire AB, the potential is equal to the potential at C? (c) If the points C and D are connected by a wire, what will be the current through it? (d) If the 4 V battery is replaced by 7.5 V battery, what would be the answers of parts (a) and (b)?



Solution: (a) Consider the adjacent figure, potential of A must be 6 and since no current is drawn from 4V battery in open circuit, potential of C = 6 − 4 = 2 V Ans.

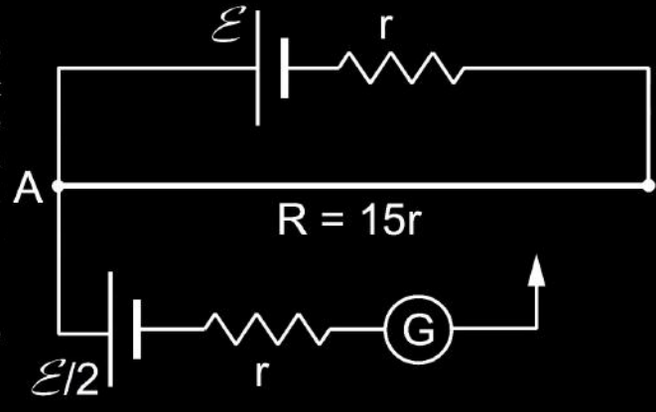


(b) Now in series potential drops are in proportion to resistance which in turn is proportional to length. Now since 6V drop occurs in a length of 100 cm, thus for a drop of 2V, required length is 100/3 i.e 33.33 cm from B or 66.67 cm from A. Ans. (c) Now if C and D are connected, no current will flow through it because they are anyway equipotential.

(d) If 4V battery is replaced by 7.5 V then $V_A = 6$ V and $V_C = 6 - 7.5$ i.e -1.5 V, now maximum drop on AB is 6V so we will not be able to locate a null point anywhere AB.

Example: H.C. Verma 56

Consider the potentiometer circuit arranged as in figure. The potentiometer wire is 600 cm long. (a) At what distance from the point A should the jockey touch the wire to get zero deflection in the galvanometer? (b) If the jockey touches the wire at a distance of 560 cm from A, what will be the current in the galvanometer?



Solution: (a) Drop across wire $V = (15/16)\mathcal{E}$ ⇒ Potential drop for a general length l: $\frac{l}{600} \times \frac{15}{16}\mathcal{E} = \frac{\mathcal{E}}{2} \Rightarrow l = 320$ cm Ans.

(b) For 560 cm, resistance of wire: $\frac{560}{600} \times 15r = 14r$. (remaining is r)

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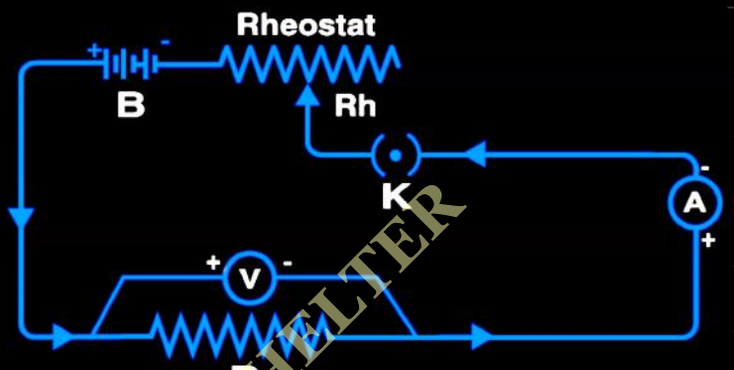
Using millman (considering lower branch as driven),

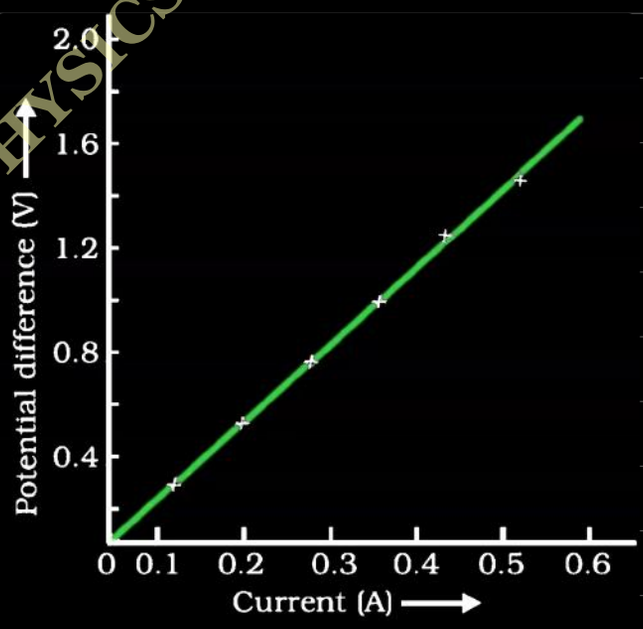
$$\mathcal{E}_{eff} = \frac{\mathcal{E}/2r + 0/14r}{1/2r + 1/14r} = 7\mathcal{E}/8 \;,\; r_{eff} = \left(\frac{2\times 14}{2+14}\right)r = 7r/4$$

Thus required current: $(7\mathcal{E}/8 - \mathcal{E}/2) / \left(\frac{7}{4}r + r\right) = 3\mathcal{E}/22r$ Amps.

VERIFICATION OF OHM'S LAW:

Procedure: Initially we close the key K and keep the rheostat resistance high to get a small current. Now Rheostat resistance is gradually decreased in steps and we take readings for current and voltage at every step and a graph between voltage and current is plotted. It is observed that this leads to a straight line graph and the resistance is calculated as slope of this graph.



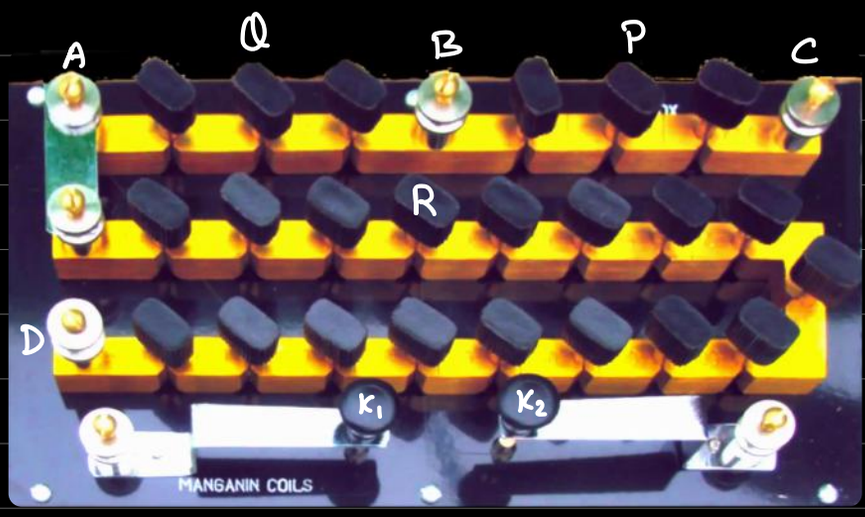


→ Note that this circuit overestimates the current since a part of current will also branch out in to voltmeter thus estimated resistance is on lower side.
(What we get is actually the parallel effection i.e $RR_v/(R+R_v)$ where R_v is resistance of voltmeter.

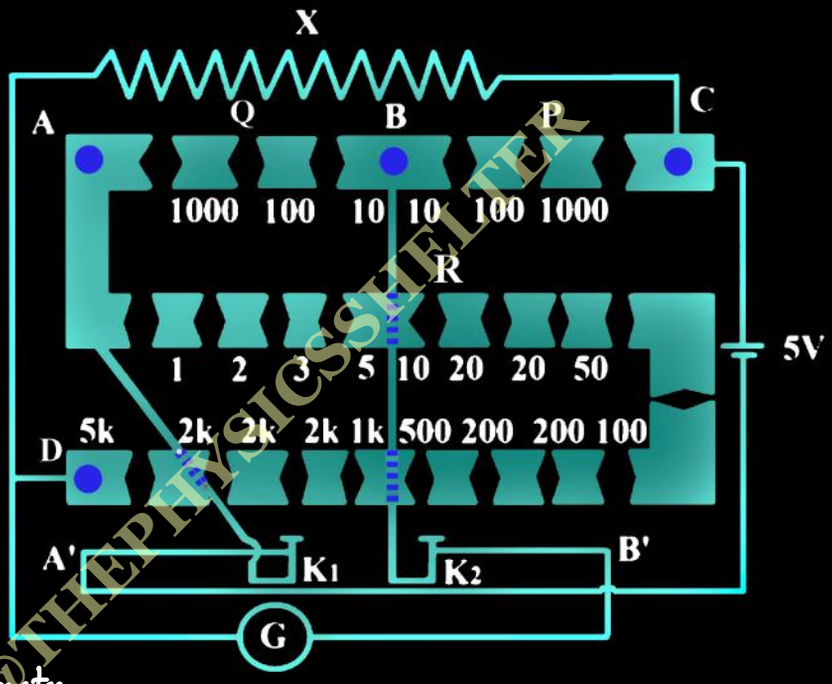
→ An alternate circuit can be arranged by placing ammeter immediately after the resistor, this gives the correct current but higher estimate of potential difference thus it overestimates the resistance, (it will give $R+R_A$ where R_A is ammeter resistance).

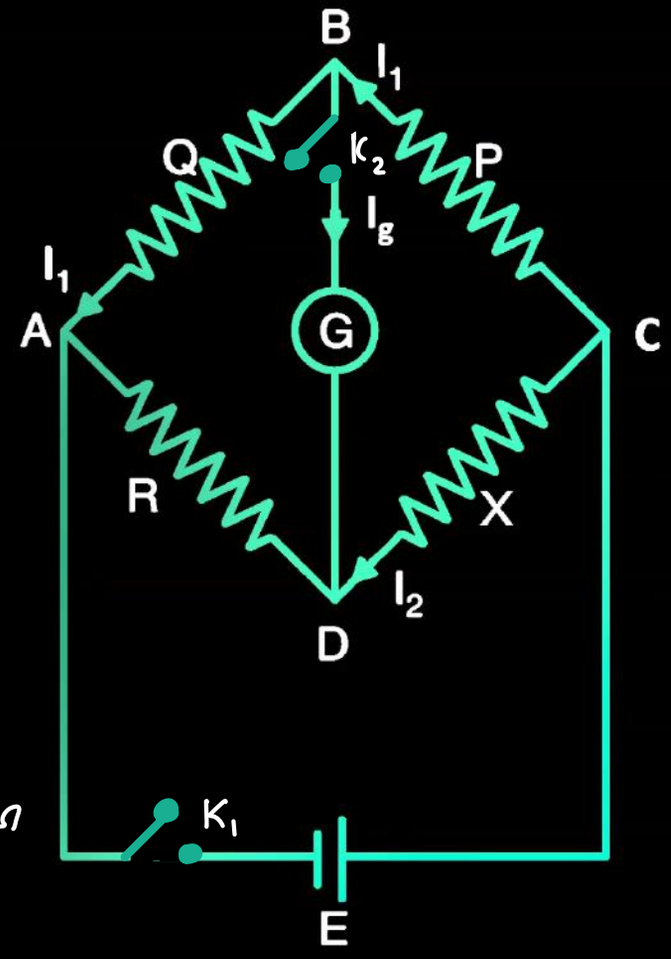
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Post Office Box:



Adjacent picture shows a post office box which can be used for finding out resistance of an unknown resistor using the principle of wheatstone bridge. It is provided with several resistances, which can be activated in a bridge circuit by pulling out several keys. The next figure shows the schematic of the instrument with connections. Here P and Q are called ratio arms, X is the unknown resistance and R can be varied in steps of 1Ω. The third figure shows equivalent circuit diagram. For galvanometer to show zero deflection, $P/Q = X/R$ $\Rightarrow X = (P/Q)R$. However since R is being varied in discrete steps of 1Ω, we will not get zero deflection however deflection will change its direction as R is varied from a value less than that required for balancing to a value more than that required for balancing. Suppose our unknown resistance were 3.1415Ω, then we follow the following steps:





① First keep $P/Q = 1$, thus at $R = 3\Omega$ deflection will be in one direction and at $R = 4\Omega$, the deflection

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will be in reverse direction. Thus we get to know the resistance correctly up to units digit, i.e it is $3 \cdot \underline{x}\ \underline{x}\ \underline{x}\ \underline{x}$.

② Now we make P/Q = 1/10. Thus now balancing value of R is 31.14 which means deflection will reverse as R is changed from 31Ω to 32Ω. Thus we estimate first decimal place of resiston as

$X = 3 \cdot 1\ \underline{x}\ \underline{x}\ \underline{x}$

③ Now we make P/Q = 1/100 thus balancing value of R is 314.5 Ω thus reversal of direction of deflection of Galvanometer is obtained when R is increased from 314Ω to 315 Ω, thus we know that our required resistance is $X = 3 \cdot 14__$

④ Now theoretically one can also reduce P/Q to 1/1000 and 1/10,000 to evaluate third and fourth decimal place of the unknown resistance.

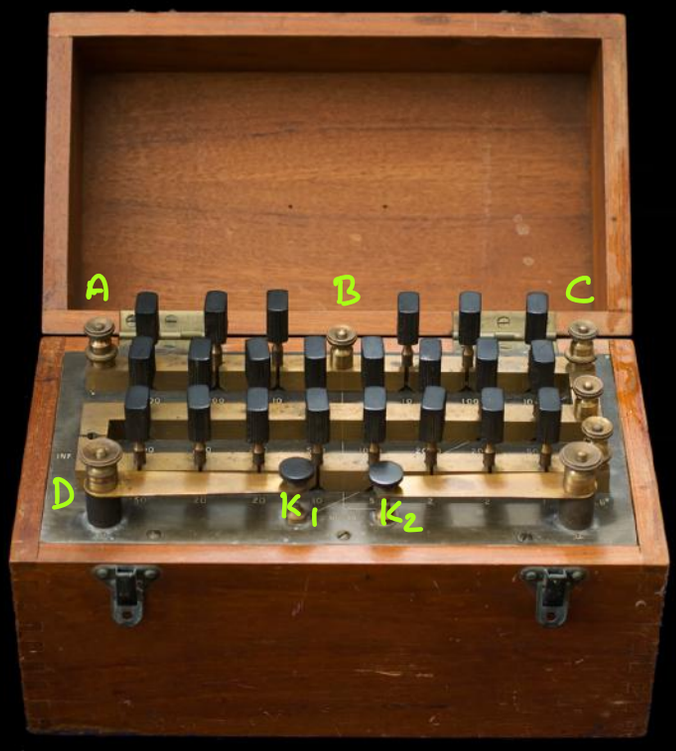
→ For largen resistance we can also start with P/Q = 10, 100 or 1000 also. (To save time).

→ In most common post office boxes, 1/1000 and 1/10,000 options are not available, in a typical post office box, P/Q may be varies from 100 to 1/100.

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EXAMPLE: Consider the Post office box apparatus shown in the adjacent figure, identify the pair of points between which we connect:

(a) Unknown Resistance
(b) Galvanometer
(c) Battery
(d) Which key activates battery
(e) Which key activates Galvanometer.



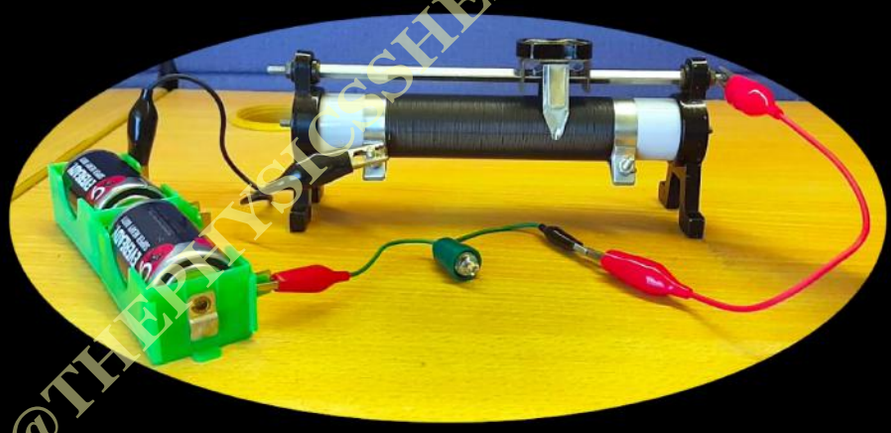
Solution: (a) Between C & D
(b) Between B & D (Through K_2)
(c) Between A & C (Through K_1)
(d) K_1
(e) K_2 [Refer to the schematic figure, also note the

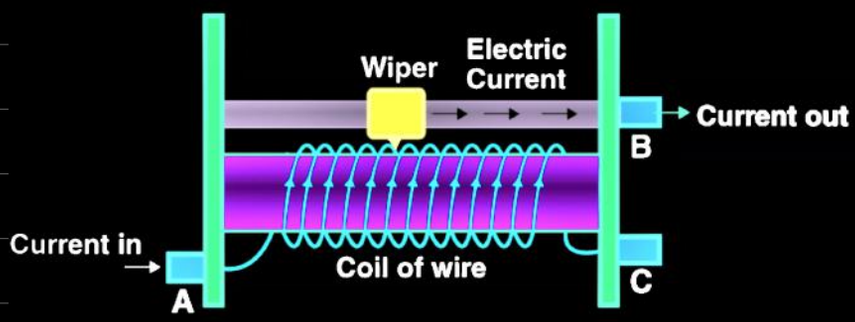
scratch on metal plate to show which key is connected to which point through internal connection in the P.O. Box.

Resistance Box: It is a box in which several resistances which can be activated by pulling out various keys (The keys are shorting the resistors between two highly conducting thick copper conductors) The arrangement can be seen in the adjacent figure.



Rheostat: A rheostat is essentially a resistance coil from which we can be obtain desired value of resistance by suitably sliding a "wiper" as shown in the figure. It has two fixed terminals and one sliding terminal. For using Rheostat to provide variable resistance only two terminals are used (sliding and one fixed). It can be also used as a potential divider by connecting the fixed terminals across a battery and getting desired voltage by setting the slides.



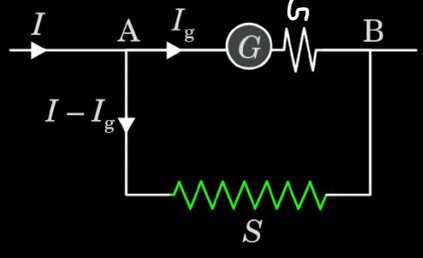


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Example : Pathfinder Build-28

A multi-range milli-ammeter is made by providing a set of interchangeable shunts and a microampere range galvanometer. This milli-ammeter is connected in a circuit to measure current. When 1.0 mA range is selected, it reads $I_1 = 1.0$ mA and when 3.0 mA range is selected it reads $I_2 = 1.5$ mA. What is the true current I_0 in the circuit without the milli-ammeter?

Solution: The range is being increased by a factor of the order of 10^3 thus practically shunt itself is the effective resistance of milliammeter. Let $\mathcal{E}_H$ and R_H be the thevenin parameters across the ammeter and let the shunts for 1mA and 3mA range be 3S and S respectively.



Thus:

$$\frac{\mathcal{E}_H}{R_H + 3S} = 1\,mA \quad (1) \quad \text{and} \quad \frac{\mathcal{E}_H}{R_H + S} = 1.5\,mA \quad (2)$$

and we need to find $\mathcal{E}_H / R_H$, i.e $I_0 = \frac{\mathcal{E}_H}{R_H}$ (3)

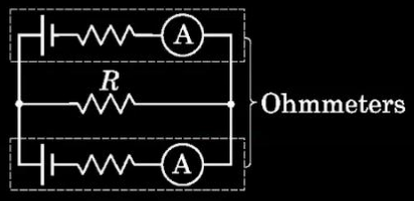
Solving (1), (2) & (3) consider, I_0, S and R_H as unknowns:

$$I_0 = 2mA \;, \quad R_H = 500\,\mathcal{E}_H, \quad S = \frac{500\,\mathcal{E}_H}{3}$$

Thus $I_0 = 2\,mA$ ohms.

EXAMPLE: PATHFINDER: BUILD-29:

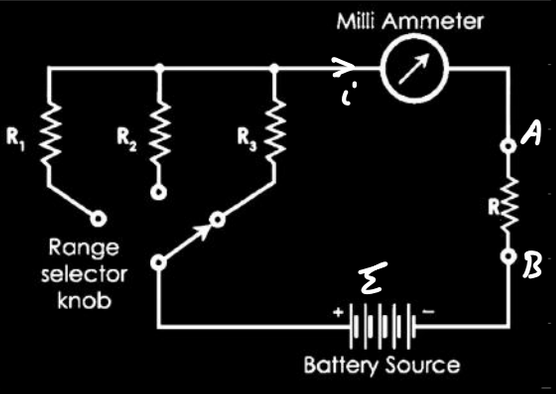
A homemade Ohmmeter consists of a series connected battery, a resistance and an ammeter. A student constructs two such ohmmeters using non-identical components and calibrates them by measuring the same set of known resistances. These two Ohmmeters are together connected in parallel with an unknown resistance R as shown in the figure. If one of the ohmmeter reads $R_1 = 30\ \Omega$ and other reads $R_2 = 60\ \Omega$, find true value of the unknown resistance R.



Solution: Let us first understand calibration of ohm meter. Consider the adjacent figure.

Here $i = \frac{\mathcal{E}}{r + R}$ where r is the series resistance of the loop excluding the load resistance. Thus, $R = \frac{\mathcal{E}}{i} - r$ or

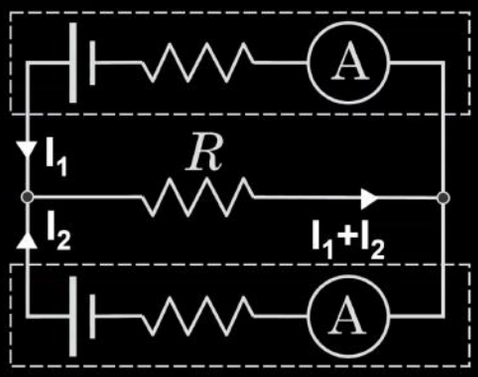
$$R = \frac{\mathcal{E} - ir}{i} = \frac{V}{i}$$ where V is the voltage drop between points A and B. Thus ohm meter measures the ratio of potential drop across its terminals to the current through its ammeter. Now let us use this information in our current problem.



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Consider the adjacent figure. Let V be the potential drop across the resistor R then according to our question:

$$\frac{V}{I_1} = R_1 \ (1) \qquad \frac{V}{I_2} = R_2 \ (2)$$



and $$R = \frac{V}{I_1+I_2} = \frac{V}{(V/R_1)+(V/R_2)} = \frac{R_1R_2}{R_1+R_2}$$

Thus required resistance, $R = \frac{R_1R_2}{R_1+R_2} = \frac{30\times 60}{30+60}\ \Omega = 20\ \Omega$ Ans.

EXAMPLE: PATHFINDER: MCQ 12:

An ideal battery of electromotive force $\mathscr{E}$ is connected in series with an ammeter and a voltmeter of unknown internal resistances. If a certain resistance is connected in parallel with the voltmeter, the voltmeter and the ammeter readings becomes $1/\eta$ and η times of their respective initial readings. What is the initial reading of the voltmeter?

(a) $\frac{\mathscr{E}}{(\eta+1)}$ (b) $\frac{\eta\mathscr{E}}{(\eta+1)}$

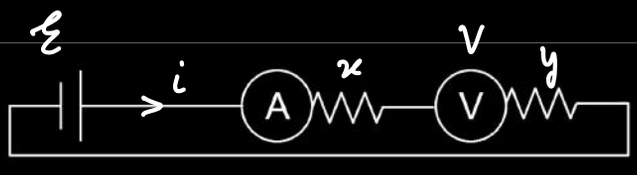
(c) $\frac{\eta\mathscr{E}}{(\eta-1)}$ (d) $\frac{(\eta+1)\mathscr{E}}{\eta}$

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Solution: Applying loop law:

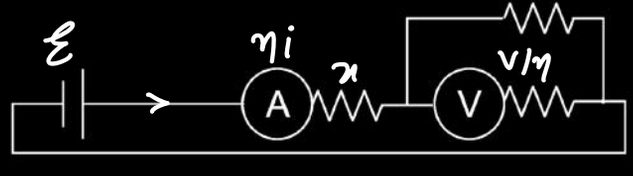
Initial circuit: $\mathscr{E} = ix + V$ (1)

Final circuit: $\mathscr{E} = \eta ix + V/\eta$ (2)



Now treating ix and V as unknowns in equations and solving we get:

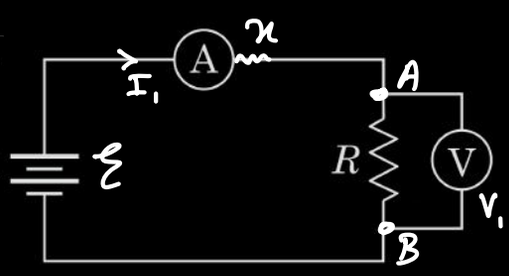
$$ix = \frac{\mathscr{E}}{1+\eta}, \quad V = \eta\mathscr{E}/\eta+1$$



Thus $V = \eta\mathscr{E}/(\eta+1)$ i.e (b) Ans.

EXAMPLE: PATHFINDER: BUILD:25:

In the circuit shown, the voltmeter reads $V_1 = 1.0$ V and the ammeter reads $I_1 = 1.0$ A. On changing the places of the ammeter and the voltmeter, the voltmeter will read $V_2 = 2.0$ V and the ammeter will read $I_2 = 0.5$ A. Find resistance R, resistances R_V of the voltmeter and R_A of the ammeter.

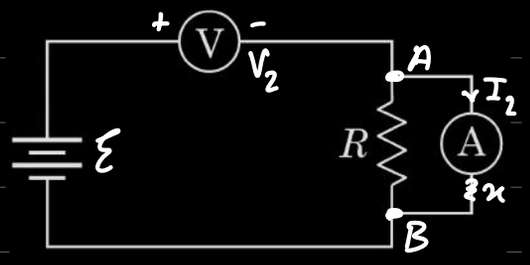


Solution: Let ammeter resistance be x. Then

from first circuit:

$$\mathcal{E} = I_1 x + V_1 \quad (1)$$

From second circuit: $\mathcal{E} = I_2 x + V_2$ (2)



Solving we get:

$$\mathcal{E} = \frac{V_2 I_1 - V_1 I_2}{I_1 - I_2} \quad \text{and} \quad x = \frac{V_2 - V_1}{I_1 - I_2} = R_A \quad \text{Ans.}$$

Now from first loop:

$$I_1 = \frac{V_1}{R_v} + \frac{V_1}{R} \quad (3) \quad (\because \text{Junction law at joint A})$$

From Second loop: $\frac{V_2}{R_V} = I_2 + \frac{I_2 x}{R}$ (4) ($\because$ Junction law at joint A)

Solving for R and R_v from equations (3) & (4) and substituting x, we get:

$$R = \frac{V_1}{i_1 - i_2}, \quad R_v = \frac{V_1}{I_2} \quad \text{Ans.}$$

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